Evolving embodied intelligence

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Introduction to AI Course 2022, VU, 26-09-2022

1

2

Evolution can create intelligence

Artificial evolution can create artificial intelligence

3

Can we evolve intelligence?

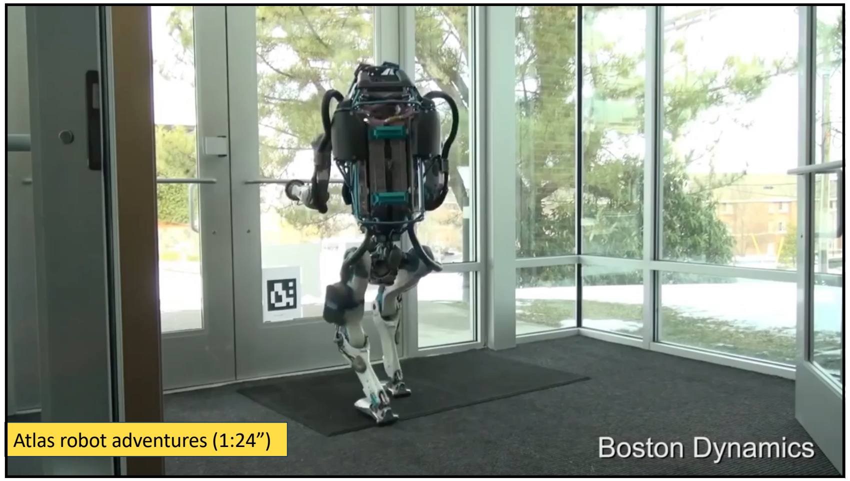
4

AI

Human

Google Duplex Makes Appointment (0:58”)

5



6

Digital AI

Embodied AI

machine translation

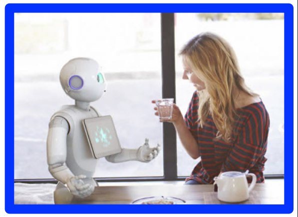
robot receptionist

chatbots

medical diagnosis

warehouse robots

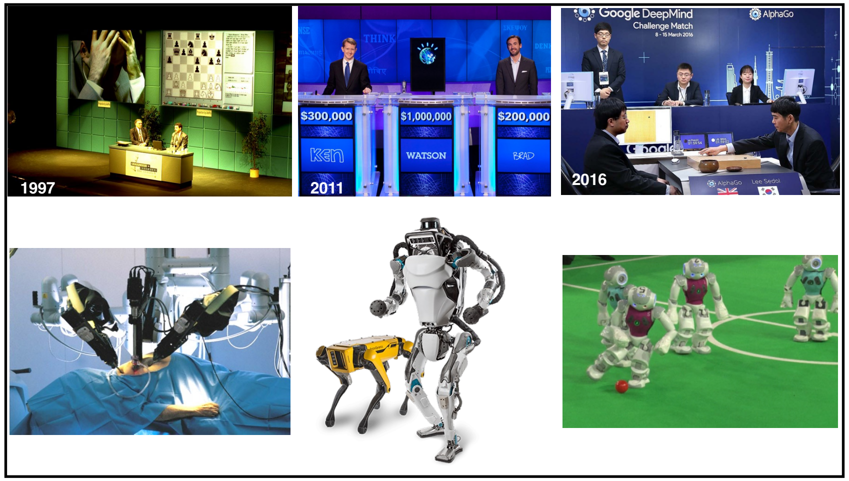
self-driving cars



Thinking machines

Acting machines

7



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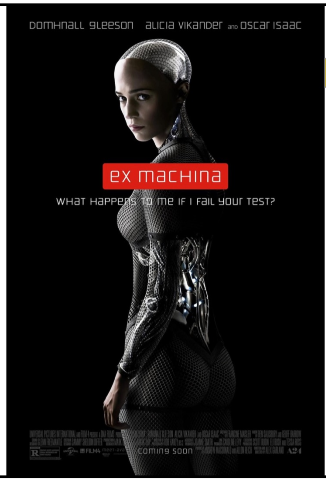

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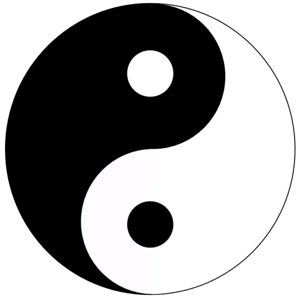
9

Body

Hardware



Brain

Software

10

$$\text{Environment} + \underbrace{\text{body} + \overbrace{\text{brain}}^{\text{intelligence (narrow)}}}_{\text{intelligence (broad)}} \rightarrow \text{behavior}$$

11

Optimal body = ?

Pragmatist

Antropo-chauvinist

Radicalist

Optimal brain = ?

12

13

Can we evolve embodied intelligence?

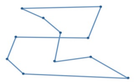
Can we evolve robots?

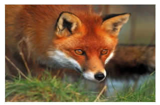
14

Artificial evolution?

15

Individuals



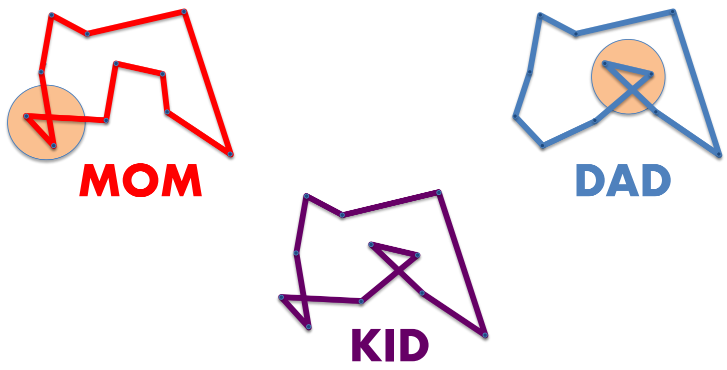


Natural selection

Reproduction

digital sex

16



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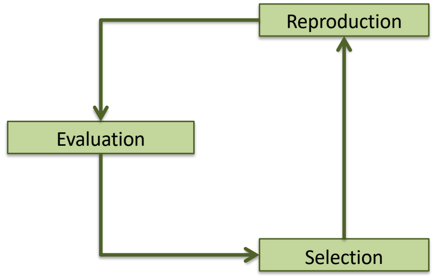
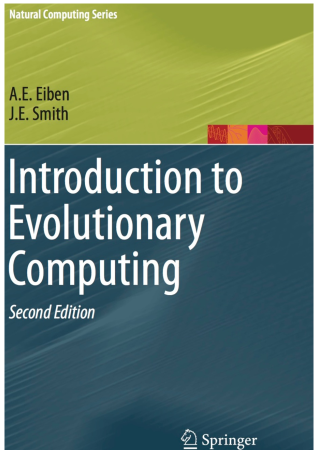


18



Genetic algorithms

Evolution strategies

Evolutionary programming

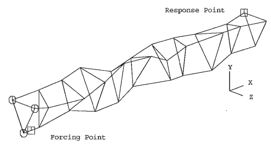
Genetic programming

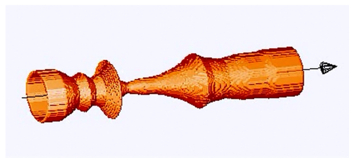
Differential evolution

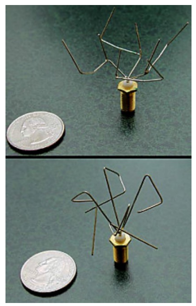
...

19

Evolutionary design







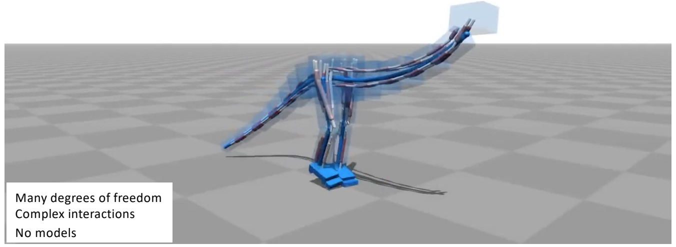
20

passive ⟷ active

agency
autonomy

21

Geijtenbeek et al., Flexible Muscle-Based Locomotion for Bipedal Creatures, ACM ToG, 2013



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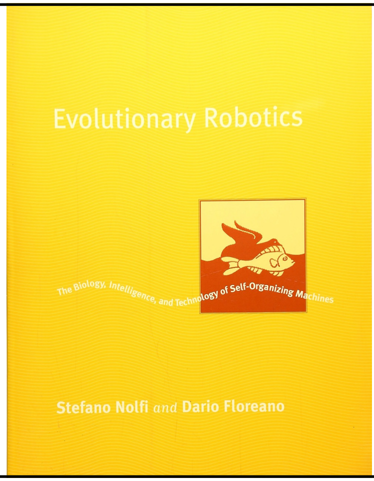
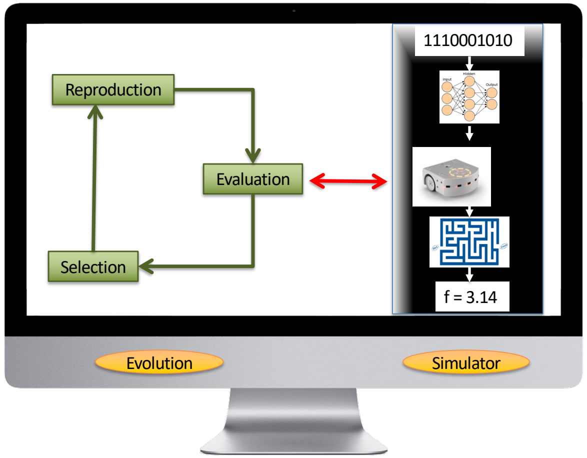


23

2[nd] order engineering

"Aims to apply evolutionary computation techniques to evolve the overall design or controllers, or both, for real and simulated autonomous robots" (Vargas et al., 2014)"

24



25

Matter matters

REALITY GAP

26

real

sweet spot

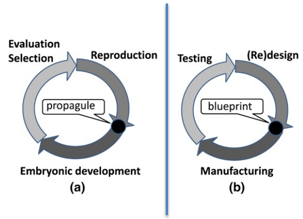
passive

active

virtual

27

Eiben, Kernbach, Haasdijk, Embodied artificial evolution: Artificial evolutionary systems in the 21st Century, Evolutionary Intelligence, 5(4):261-272, 2012



GRAND CHALLENGES

1. Body types
2. How to start
3. How to stop (kill switch)
4. Evolvability & speed
5. Process control
6. Body+brain evolution + learning

28

Eiben and Smith, From evolutionary computation to the evolution of things, ***Nature***, 2015

20th century

21st century

WETWARE
in vivo

SOFTWARE
in silico

HARDWARE
in materio

Biosphere

Evolutionary Computing

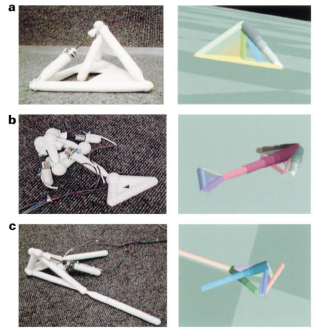
Evolution of Things

29

Lipson, Pollack, Automatic design and manufacture of robotic lifeforms, *Nature* 406, 2000

- GOLEM project
- EA + simulator
- Evolution of body + controller
- Evolved robot fabricated afterwards

- Off-line evolution
- Evolution in simulation
- No sensors – are they robots?



30

Watson, Ficici, Pollack, Embodied Evolution, *Robotics and Autonomous Systems* 39, 2002

- Evolution in physical robot population
- On-line distributed EA for evolving controllers

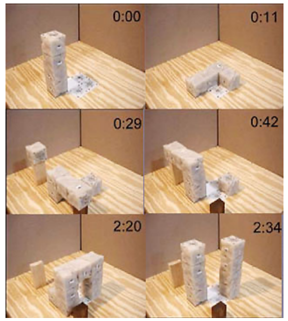
- Bodies are fixed

31

Zykov, Mytilinaios, Adams, Lipson, Self-reproducing machines, *Nature* 435, 2005

- Physical system with Molecubes
- Raw material manually replenished
- Robot <u>**constructs a replica**</u> by lifting and assembling cubes from the feeding locations

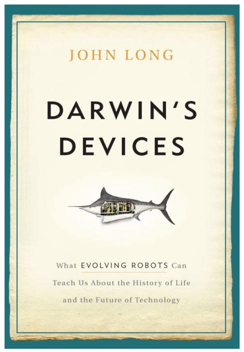
- Not evolvable (no genotype)
- 2nd robot is exact clone
- “Self…” is a bad idea, cf ethics later



32

John Long, Darwin's Devices: What Evolving Robots Can Teach Us About the History of Life and the Future of Technology, *Basic Books* 2012

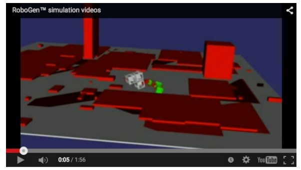
- Studies a biological question
- Real robots form **physical model** of a biological phenomenon
- All fitness evaluations are in hardware
- It really teaches us something (no spoiler here)
- It takes LONG ☺

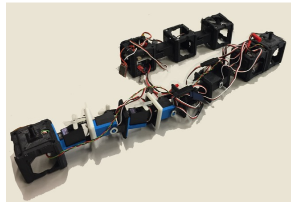


33

Auerbach *et al.*, RoboGen: Robot generation through artificial evolution, Proc. of Artificial Life 2014, MIT Press, 2014

- Didactic tool and experimental testbed
- Open-source SW & HW platform
- Robots:
  - Fixed parts, 3D-printed parts
  - Arduino board, NN controllers
  - Tree-based genetic code
- Evolution
  - Off-line, centralized, supervised, simulated
  - Fitness evaluation in isolation
- End product of simulated evolution is constructed in HW by hand

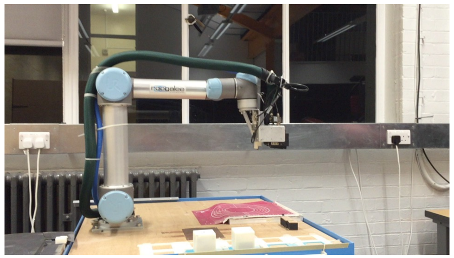




34

Brodbeck, Hauser, and Iida, Morphological evolution of physical robots through model-free phenotype development, PLoS One, 10(6) 2015.

- Evolution
  - Off-line, centralized, physical trials
  - Fitness evaluation in isolation
  - Only mutation, no crossover
- Each individual is constructed in HW
  - By "mother robot"
  - Hands-free in Experiment 1
  - Human-assisted in Experiment 2
- No sensors – are they robots?



35

Jelisavcic, De Carlo, Hupkes, Eustratiadis, Orlowski, Haasdijk, Auerbach, Eiben, Real-World Evolution of Robot Morphologies: A Proof of Concept, ***Artificial Life 23(2),*** 2017.

Full movie (5min): https://www.youtube.com/watch?v=BfcVSb-Q8ns

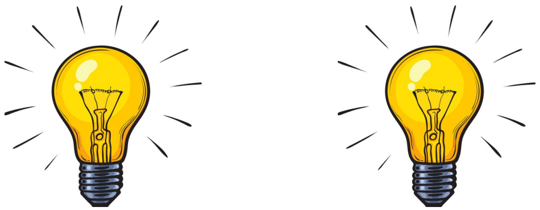
36

Can we evolve real robots?

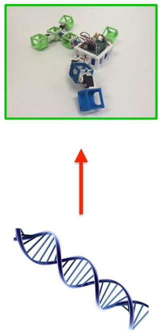
37

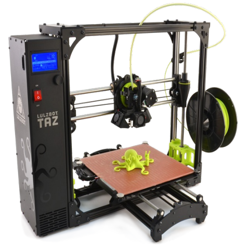
This is a good moment for a break

38



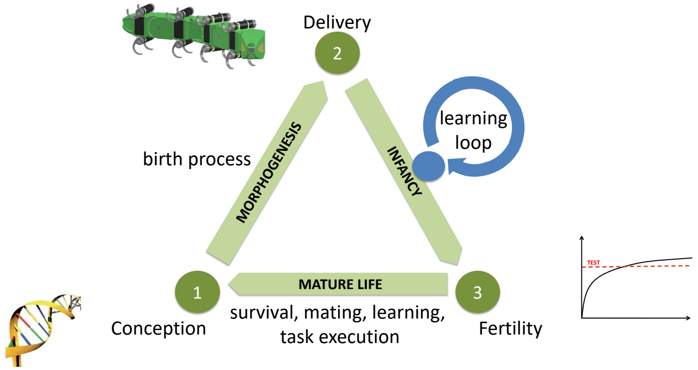
39





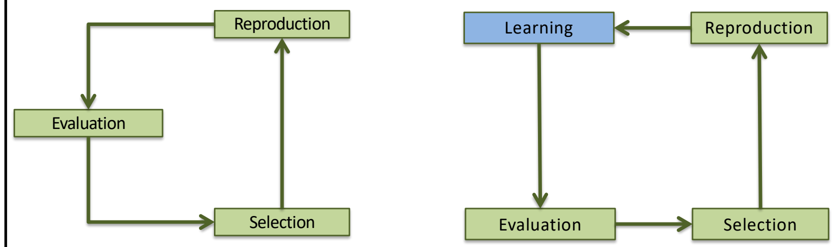
40

Eiben, Bredeche, Hoogendoorn, Stradner, Timmis, Tyrrell, Winfield, The Triangle of Life: Evolving Robots in Real-time and Real-space, ***ECAL***, 2013



41

Eiben and Hart, If it evolves it needs to learn, ***GECCO workshop***, 2020



42

Human assistance

		Selection	
		Y (= breeding)	**N**
Reproduction	**Y**	**Type 1**	**Type 2**
	N	**Type 3**	**Type 4**

43

Off-line evolution
before the job

On-line evolution
on the job

Design stage → Deployment → **Operational stage**

Breeding scenario
optimization

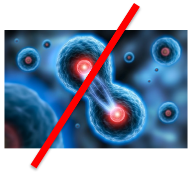
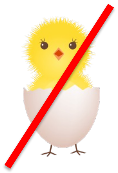
Mars scenario
adaptation

44

45

Should we evolve real robots?

46


47

Fundamental insight:
not necessary

Ethical constraint:
not desirable

~~Distributed (self-)reproduction~~

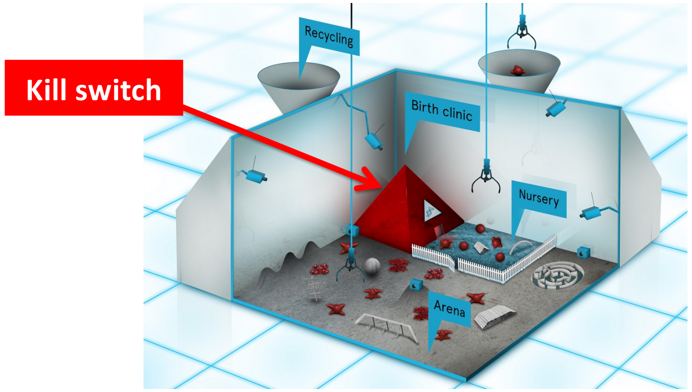
48

Centralized, externalized reproduction

Does not invalidate the concept of evolution

49

Eiben, EvoSphere: the world of robot evolution, ***TPNC conference***, 2015.

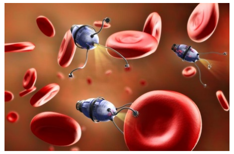


50

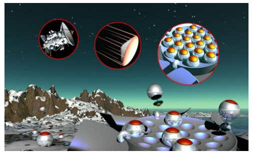
Applications

Rain forest monitoring

Seafloor mining



Medical nano robots



Terraforming

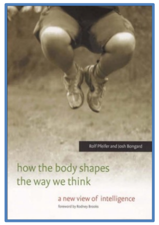
51

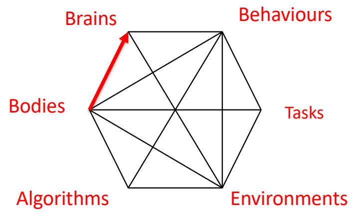
2nd order software / hardware engineering

52

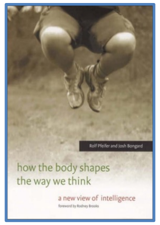
Environment + body + brain → behavior

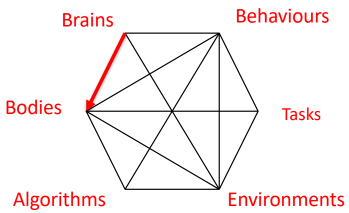
53





54





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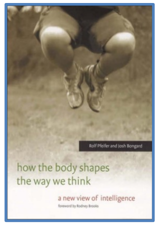
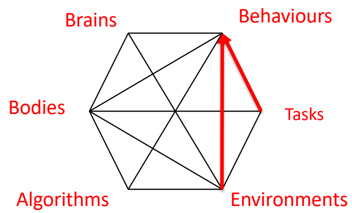
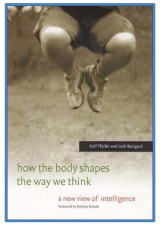
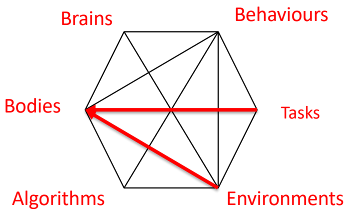




56





57

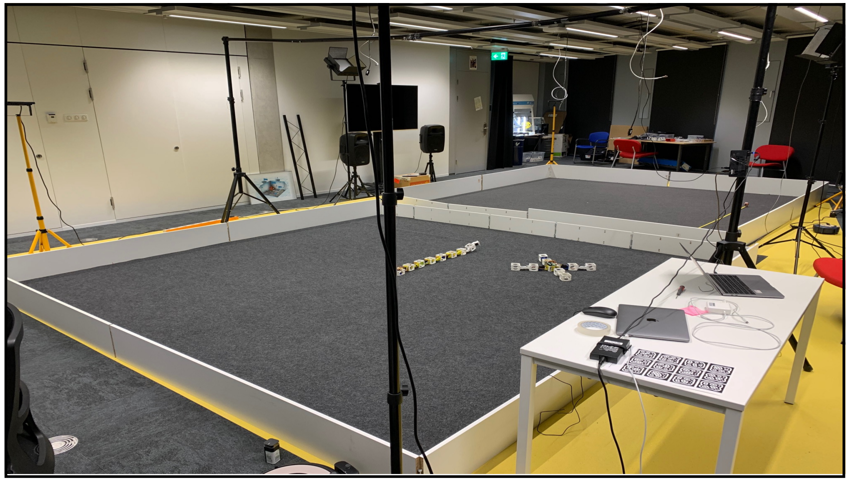
VRIJE UNIVERSITEIT AMSTERDAM

laying the groundwork

developing know-how

seeking deeper understanding
(patterns, relationships, invariants)

58



59

fit for environment vs. fit for purpose
(how to combine)

PLOS One, 2014

interaction between body and brain
(which one is "more important")

Artificial Life conf. 2018

learning and evolution
(Lamarck rules)

Frontiers in Robotics and AI, 2019

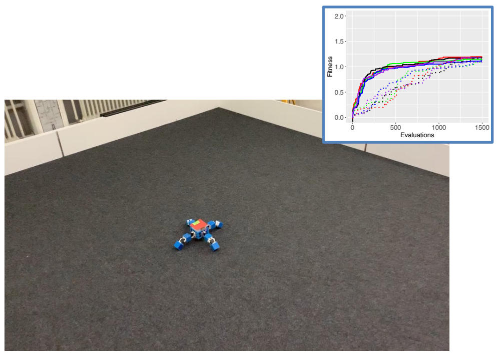
61

Evolution of intelligence

preconditions, attractors, invariants

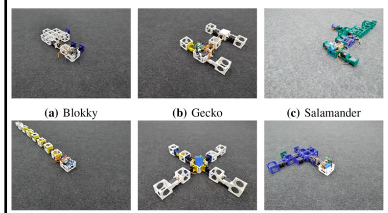
62

64



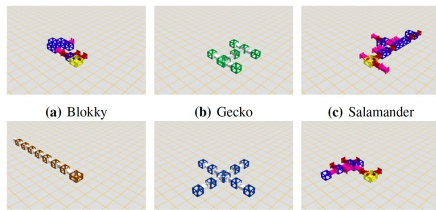
65

Machine Learning → Learning Machines



(a) Blokky (b) Gecko (c) Salamander

(d) Snake (e) Spider (f) Stingray

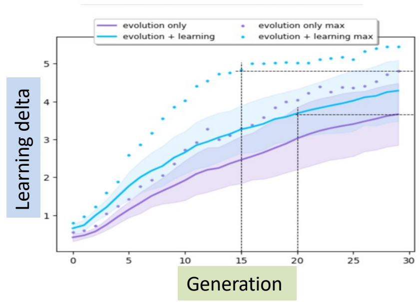


(a) Blokky (b) Gecko (c) Salamander

(d) Snake (e) Spider (f) Stingray

67

Emergence of morphological intelligence



68

Autonomous Robot Evolution project (EPSRC, 2018-2022)

World's first **real world** EvoShpere

Nuclear cave application

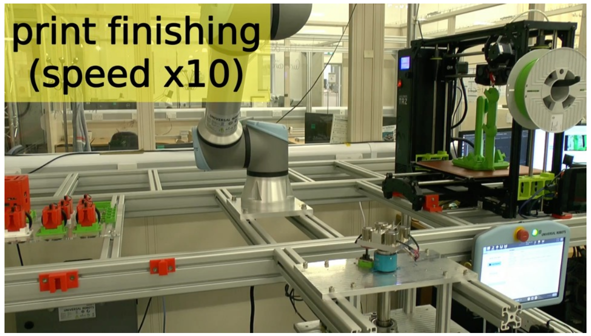
Hybrid robot evolution

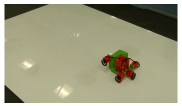
Novel hybrid evolution:

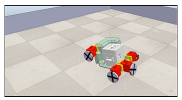
- Two populations – one species
- same genetic code
- cross-fertilization
- migration

"virtual father – physical mother"

69

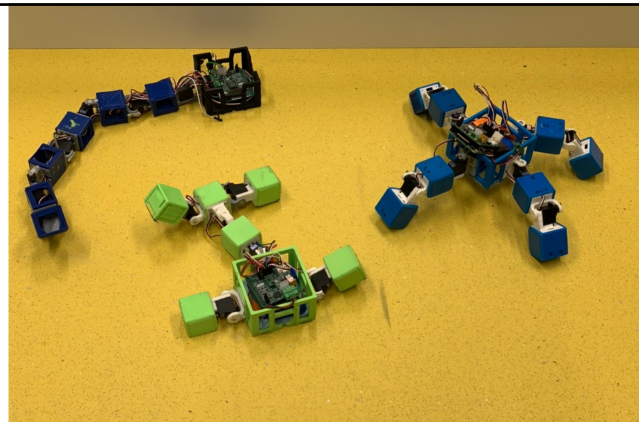






RoboFab ("Birth Clinic"), Bristol Robotics Lab

70



"robot zoo" in the bio-inspired robotics lab at the VU

71

< 5 years

Type 1+ EvoSphere, robot breeding farms

5 – 15 years: supervised

Type 2, 3 EvoSpheres

15+ years: autonomous

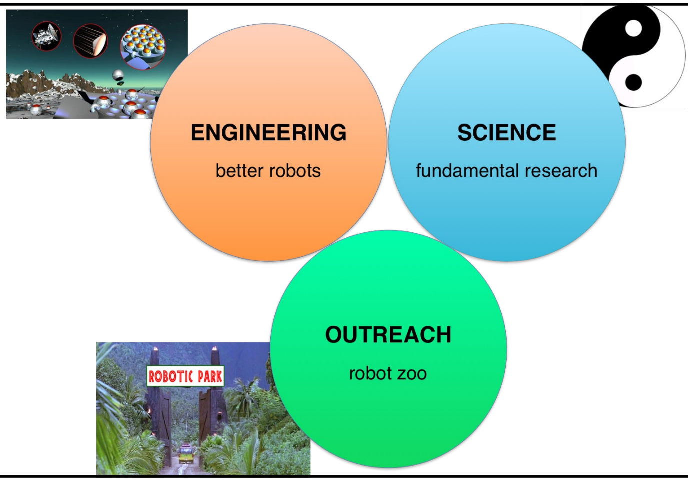
Type 4 robot colonies for mining, terraforming, entertainment, …

72

EvoSphere = novel research instrument

Stars → telescope
Particles → cyclotron
Evolution → EvoSphere

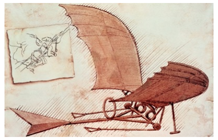
73



74

Exhibition brAIn power, Leiden, Museum Boerhave, Sept 2022 – March 2023

75



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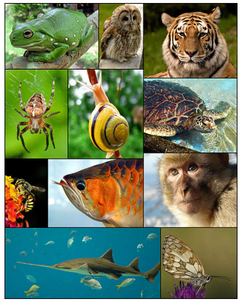
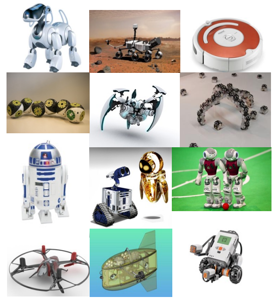
76





77

IT'S LIFE JIM

BUT NOT AS WE KNOW IT

Thanks to J. Auerbach, N. Bredeche, E. Buchanan, M. D'Angelo, M. De Carlo, F. van Diggelen, S. Doncieux , P. Eustratiadis, L. Le Goff, K. Glette, E. Haasdijk, M. Hale, E. Hart, J. Heinerman, E. Hupkes, M. Hoogendoorn, D. Howard, M. Jelisavcic, S. Kernbach, R. Kiesel, G. Lan, W. Li, J. Luo, K. Miras, J.-B. Mouret , G. Nietschke, J. Orlowski, B. Paechter, C. Rossi, J. Smith, J. Stradner, J. Timmis, A. Tyrrell, B. Weel, A. Winfield

78